

Assignment – Week 8
Introduction to Machine Learning-IIT KGP
Topic: Clustering
TYPE OF QUESTION: MCQ/MSQ
Number of Question: 10
Total Marks: 10*2 = 20

Question 1:

Which of the following statements about K-Means clustering are true?

- I. K-Means attempts to minimize within-cluster variance.
- II. K-Means always converges to the global optimum solution.
- III. The number of clusters k must be specified beforehand.

- A) I only
- B) I and III only
- C) II and III only
- D) I, II and III

Correct Answer: B

Detailed Solution: K-Means minimizes the within-cluster variance. However, it may converge to a local optimum depending on initialization, so statement II is false. The number of clusters k must be specified beforehand.

Question 2:

Suppose the following points belong to a cluster:

(2,3), (4,5), (6,7), (8,9)

What is the centroid of the cluster?

- A) (5,6)
- B) (4,5)
- C) (6,7)
- D) (5.5,6.5)

Correct Answer: A

Detailed Solution: The centroid is computed by averaging the x and y coordinates separately.

x-coordinate = $(2+4+6+8)/4 = 5$

y-coordinate = $(3+5+7+9)/4 = 6$

Hence, centroid = (5,6).

Question 3:

In hierarchical agglomerative clustering, what happens initially?

- A) All data points belong to one cluster
- B) Each data point forms its own cluster
- C) Random clusters are initialized
- D) The centroid is calculated first

Correct Answer: B

Detailed Solution: Agglomerative clustering begins with each point as a separate cluster and repeatedly merges the closest clusters.

Question 4:

Consider the following distance matrix for 4 points (i.e. p1, p2, p3 and p4):

	p1	p2	p3	p4
p1	0	2	6	10
p2	2	0	5	9
p3	6	5	0	4
p4	10	9	4	0

Using single-link hierarchical clustering, which two points merge first?

- A) p1 and p3
- B) p2 and p4
- C) p1 and p2
- D) p3 and p4

Correct Answer: C

Detailed Solution: Single-link clustering merges the pair with the smallest distance. The smallest non-zero distance is 2 between p1 and p2.

Question 5:

Which of the following are disadvantages of K-Means clustering?

- I. Sensitive to outliers
- II. Performs poorly for non-spherical clusters
- III. Requires labeled data

- A) I only
- B) I and II only
- C) II and III only
- D) I, II and III

Correct Answer: B

Detailed Solution: K-Means is sensitive to outliers and struggles with non-spherical clusters. However, it is an unsupervised algorithm and does not require labeled data.

Question 6:

Given sets:

A = {1,2,3,4,5}

B = {3,4,5,6,7}

What is the Jaccard Index of A and B?

- A) 0.20
- B) 0.33
- C) 0.43
- D) 0.60

Correct Answer: C

Detailed Solution: Intersection = {3,4,5}, so size = 3.

Union = {1,2,3,4,5,6,7}, so size = 7.

Jaccard Index = $3/7 = 0.43$.

Question 7:

Which of the following evaluation metrics is commonly used for clustering quality?

- A) Mean Squared Error
- B) Silhouette Score
- C) Accuracy
- D) Precision

Correct Answer: B

Detailed Solution: Silhouette Score measures how well points fit within their assigned clusters compared to neighboring clusters.

Question 8:

Which of the following statements about hierarchical clustering are true?

- I. Agglomerative hierarchical clustering produces a dendrogram.
- II. Agglomerative hierarchical clustering follows a bottom-up approach.
- III. Hierarchical clustering always requires specifying the number of clusters beforehand.

- A) I only
- B) I and II only
- C) II and III only
- D) I, II and III

Correct Answer: B

Detailed Solution: Hierarchical clustering produces a dendrogram and agglomerative clustering follows a bottom-up strategy. The number of clusters need not be specified beforehand.

Question 9:

What is the primary objective function minimized in K-Means clustering?

- A) Sum of inter-cluster distances
- B) Entropy of clusters
- C) Classification error
- D) Sum of squared distances from points to centroids

Correct Answer: D

Detailed Solution: K-Means minimizes the within-cluster sum of squared distances between data points and their cluster centroids.

Question 10:

Which clustering algorithm is most sensitive to the choice of initialization?

- A) Hierarchical agglomerative clustering
- B) K-Means
- C) Hierarchical divisive clustering
- D) Single-link clustering

Correct Answer: B

Detailed Solution: K-Means strongly depends on the initial centroid positions, which can affect both convergence and final cluster quality.

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